

LESSON 3-3 • SESSION 1 • PRACTICE • FORM B

Equivalent Ratios and Tables

Learning target – I can build a ratio table by scaling both rows by the same factor. · 6.RP.A.3a

REMEMBER

- **Equivalent ratios** make the same comparison. Multiply or divide **both** terms by the **same** number.
- In a **ratio table**, each column is one pair. Find the factor, then use it on the other row.
- Adding the same number to both terms does **not** keep the ratio.

WORKED EXAMPLE

A mix uses 5 scoops of powder for every 2 cups of water. How much water for 15 scoops?

Step 1 Find the factor: $15 \div 5 = 3$.

Step 2 Multiply the water by 3 too: $2 \times 3 = 6$.

Step 3 Check: $15 : 6 \div 3 = 5 : 2$.

Answer 15 scoops go with 6 cups of water

WORDS equivalent ratios (*razones equivalentes*) · ratio table (*tabla de razones*) · scale factor (*factor de escala*)

PRACTICE

show your work · label units

1. Find the missing number.

a. $5 \times \underline{\quad} = 35$

b. $\underline{\quad} \times 4 = 25.2$

c. $2.6 \times \underline{\quad} = 20.8$

d. $\underline{\quad} \div 4 = 7$

e. $40 \div \underline{\quad} = 8$

f. $24.7 \div \underline{\quad} = 3.8$

2. **Lemonade Recipe.** A recipe uses 4 lemons for every 3 cups of syrup. Complete the table, then find the syrup for 20 lemons.

Hint Count by 4s on the top row and by 3s on the bottom row.

Lemons	4	8	12	16
Syrup (cups)	3			

Answer _____ cups for 20

3. **Granola Bars.** Granola bars need 5 cups of oats for every 2 cups of honey. Complete the ratio table.

Hint Find what 5 was multiplied by, then use it on the 2.

Oats (cups)	5	10		25
Honey (cups)	2		6	

4. **Library Books.** A library packs 36 books for every 9 carts. Scale down to complete the table.

Hint Divide both rows by the same number each time.

Books	36	12		4
Carts	9		2	

5. **Trail Mix.** A recipe uses 7 cups of nuts for every 4 cups of raisins. How many cups of raisins go with 35 cups of nuts?

Hint $35 \div 7$ gives the scale factor.

Show your work.

Answer _____ cups

6. **Markers to Crayons.** A kit has markers to crayons in a ratio of 3.5 to 7. How many crayons go with 21 markers?

Show your work.

Answer _____ crayons

7. **Bike Ride.** A cyclist rides 12.5 miles every 2.5 hours. How far does he ride in 7.5 hours?

Hint $2.5 \text{ hours} \times 3 = 7.5 \text{ hours}$.

Show your work.

Answer _____ miles

8. **Ribbons to Bows.** A craft kit uses 1.5 yards of ribbon for every 4 bows.

a. Write two equivalent ratios that use only whole numbers. _____

b. Is $3.5 : 6$ equivalent to $1.5 : 4$? Explain. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Keisha's Work.** Keisha says $3 : 5$ and $6 : 8$ are equivalent because she added 3 to each number. Explain her mistake. Then write a ratio that really is equivalent to $3 : 5$.

TEST PRACTICE

choose the best answer

10. Which ratio is equivalent to $3 : 4$?

A $6 : 7$

B $4 : 3$

C $9 : 12$

D $5 : 6$

11. A ratio table has the columns $2 : 3$ and $6 : 9$. What was done to both rows?

A added 4

B multiplied by 3

C multiplied by 6

D added 6

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

32 Z	42 E	6.5 C	20 L	6.3 S	100.8 Q	37.5 D	15 A	19 X
#1b		#1f	#2	#5	#6	#7		

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____